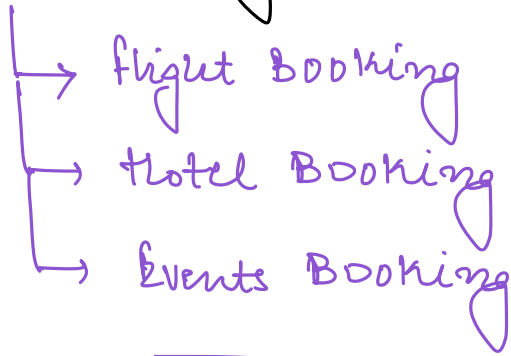


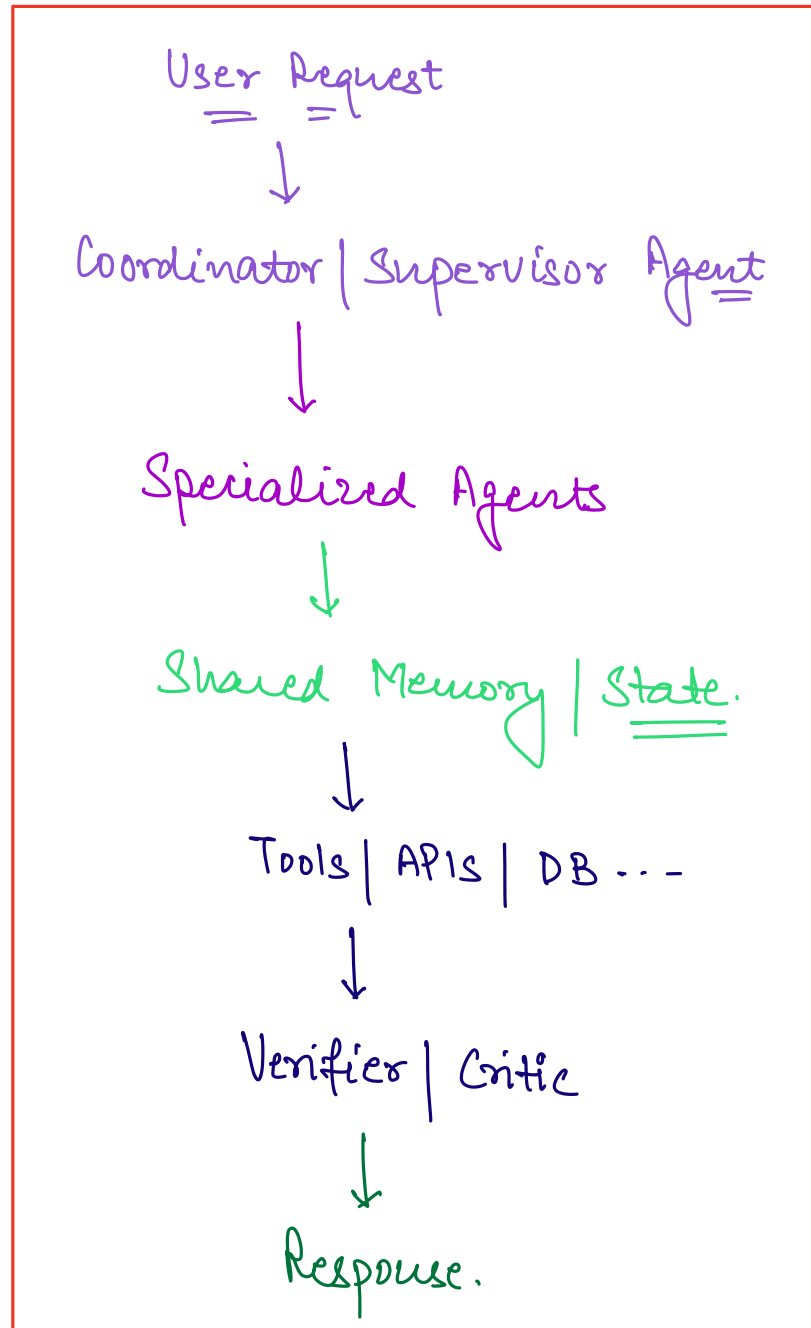
Multi-Agent architecture.

⇒ When multiple specialized agents are working together to solve a bigger goal.

Travel Agency



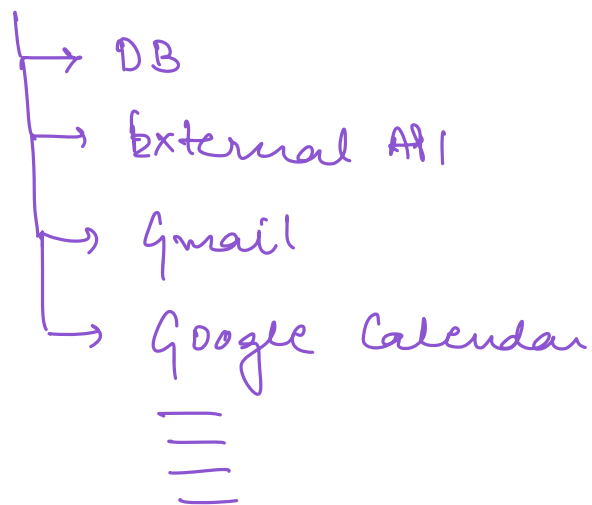
Core Components of a multi-agent architecture.



→ Agent {
Coordinator Agent
Specialized Agents

→ Shared Memory | State.

→ Tool



Multi-Agent Architecture Patterns.

→ Supervisor | Coordinator Pattern

→ Sequential Pattern

→ Hierarchical Pattern

Communication Protocols

→ Direct Message Passing

→ Hand-Off Protocol.

→ Shared Memory Protocol.

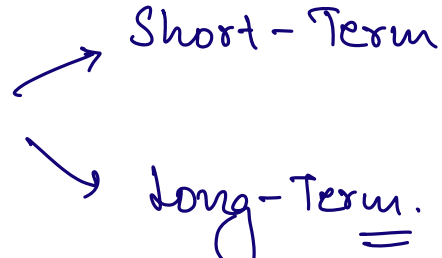
Task Delegation & Coordination

Task delegation means deciding:

- 1) What should be done?
- 2) Who should do it?
- 3) In what order?
- 4) What input is required?

⇒ Coordinator Agent.

Shared Memory | State



```
graph LR; A["# Shared Memory | State"] --> B["Short-Term"]; A --> C["Long-Term"];
```

Short-Term

Long-Term

Consensus Mechanisms.